

Lecture 31

4/8/26
JRA

Computed Tomography (CT)

Single projection leads to superposition

CT - take 1000s of slices

See BB explanation i. slide w/ cylinder, ...

Original incarnation

Pencil beam + single detector 160

Translate / rotate / translate / ... 180 angles

Did math, reconstructed attenuation coefficients as a fn of position

SLOW

Now on 3rd generation CT scanner (4th Fizzel)

Rotate / rotate scanner

5 min $\rightarrow$ seconds

Key attributes ① Fan beam of X-rays that encompasses entire patient width (so no need to translate)



(2) Use many detectors

→ Source & detectors rotate in synchrony to generate different views

(3) To get different slices thru patient



see SSCT vs MSCT
 ↑ single slice ↓ multislice

SSCT Original scanners detectors were NOT divided along $z \Rightarrow$ one slice at a time detected

Now have multi-slice CT scanners divided along z (up to 64 simultaneous slices)

intermittent (4) Patient movement along z used to be stop start (stop collect start)

continuous Now continuous movement called spiral CT

Collecting lots of data
 How do you reconstruct slices?

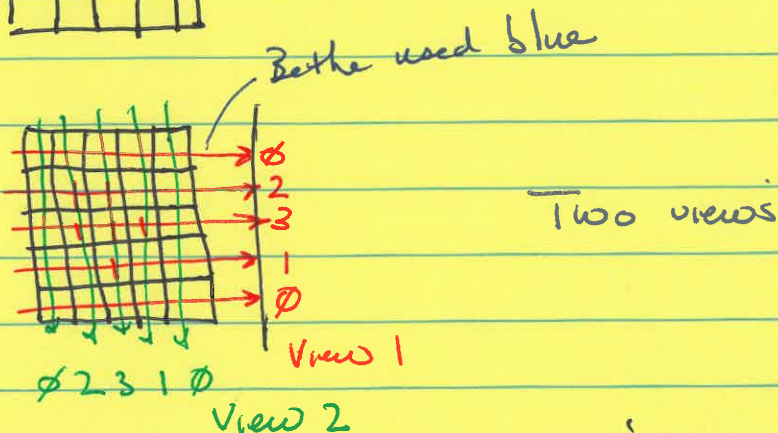
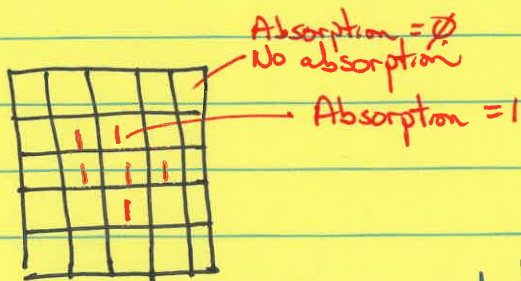
Reconstruction

Current gold std is filtered back projection (FBP)

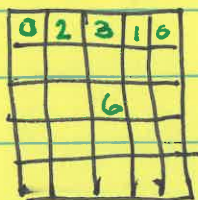
Start w/ simplified version

Start w/ simple back projection (SBP)

Doctor makes decision about resolution



Back projection - smear back ^{ε add} attenuations



attenuating
Bulk of material is
where it's supposed to be.
central

Looked at ^{PP} slide w/ phantoms

Smear & add

Smear Forward
to get projection
smear back & add

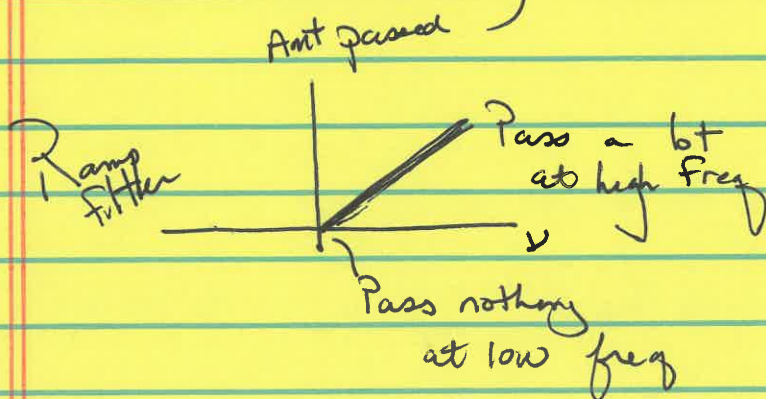
- (1) Bulk of ^{atten} material in the correct location
- (2) Blur exists (plagued by blur)

Filtered back projection eliminates blur

Is blur low-frequency or high-frequency?
coarse details fine details

Blur is excess low-frequency information in the image, so want a filter that removes low-freq info

Most rudimentary filter is the ramp filter



SUMMARY

Strengths - Fast, cheap
 Excellent images of skeleton
 Excellent soft tissue contrast using CT
 (or contrast agent)
 Anyone can have a CT scan
 but not an MRI
 Less sensitive to movement

Weaknesses Ionizing radiation →
 It NOT the most sensitive measure
 of fcn
 Better ways to get anatomical
 information

MOVIES

COVID movies w/ lung damage

Great anatomical technique

Medicine
Nuclear Imaging
 Radionuclide Imaging

Often used w/
 X-ray techniques

X-Ray

(1) Transmission-based
(X-rays transmitted thru body)

(2) Anatomical info

(3) Excellent resolution
(0.1 mm - 1 mm)
not diffraction limit

(4) Poorer contrast

(5) Less sensitive

Nuclear

(1) Emission-based
(gamma rays emitted from body)

(2) Functional info

(3) Relatively poor res
5mm - 3cm

(4) Excellent ^{intrinsic} contrast
(bkgd $\approx$ zero)
signal against dk bkgd

* (5) Extremely sensitive

Doses in nuclear medicine are very low, so

Sensitivity is excellent

Required doses extremely low

Prob of perturbing fcn w/ what you add
is correspondingly low

⇒ Good way to assay fcn

Contrast agents in X-ray & MRI may cause
issues simultaneous - A to F

Name JRA

Physics 390 In-Class Activity 11

(DO NOT INCLUDE SOURCES WHEN YOU BACK PROJECT)

OBJECT						RECONSTRUCTION							
					P1						P1		
0	0	0	0	0	∅	0	0	1	14	0	1	0	∅
0	0	0	0	0	∅	0	0	1	14	0	1	0	∅
0	0	12	0	0	12	12	12	13	26	13	12	12	12
0	1	2	1	0	4	4	4	5	18	4	5	4	4
0	0	0	0	0	∅	0	0	1	14	0	1	0	∅
∅	1	14	1	∅	P2	∅	1	14	1	∅			P2

Red # = Blue # + Green #

Questions

1. The bulk of the attenuating material is in the center of the object. Is this also true for the image produced by this very rudimentary simple back projection?

Yes - the bulk of the attenuating material is in the center of the reconstruction

2. Does this back projection suffer from blur (i.e., attenuating material appearing in incorrect parts of the image)?

Yes

3. Do you think that medicine can do better?

Yes

Name _____

Physics 390 In-Class Activity 11

(DO NOT INCLUDE SOURCES WHEN YOU BACK PROJECT)

OBJECT					RECONSTRUCTION				
					P1				P1
0	0	0	0	0					
0	0	0	0	0					
0	0	12	0	0					
0	1	2	1	0					
0	0	0	0	0					
					P2				P2

Questions

1. The bulk of the attenuating material is in the center of the object. Is this also true for the image produced by this very rudimentary simple back projection?
2. Does this back projection suffer from blur (i.e., attenuating material appearing in incorrect parts of the image)?
3. Do you think that medicine can do better?